

08 Video as World Simulator

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12 MIN READ · INTERACTIVE

Add an action input to a video model and it becomes steerable in principle. How to tell conditioning from control, what scaling bought, and why fitting physics is not the same as having the rule.

Take a model that predicts the next frame of video, and add one input.

Alongside the frames it has already seen, tell it which key is being held. The model is now capable of giving the same opening one continuation for left and another for right. That action input is necessary for an interactive world. It does not, by itself, prove the model obeys it.

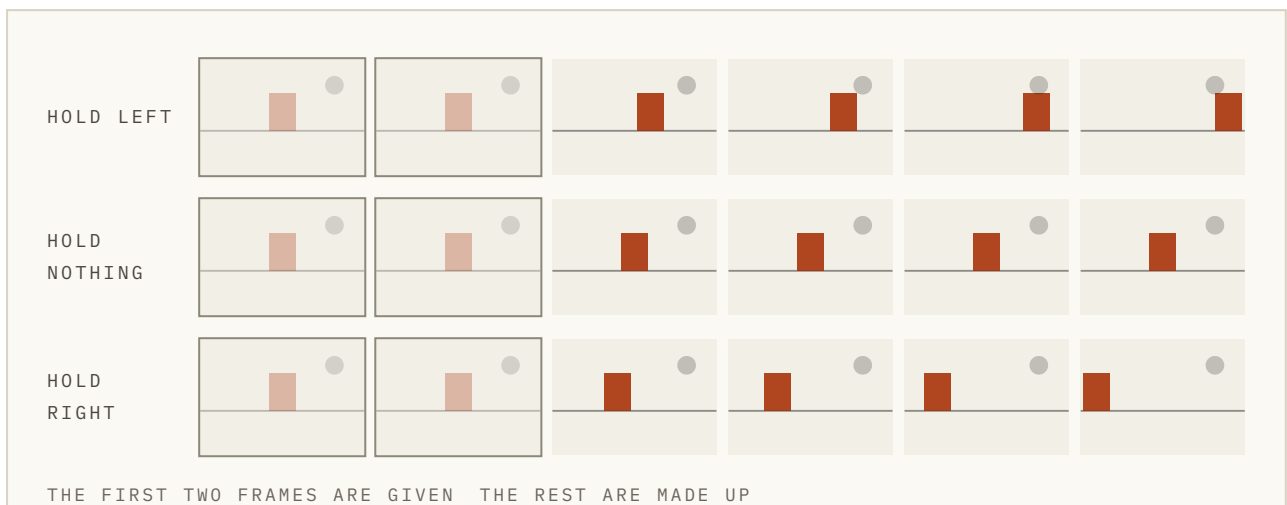


FIG. 8.1 One model, one start, three futures. The first two frames are given; the rest are made up. Turn the action input off and all three rows become identical. Without it there is one future on offer, whichever the model finds most likely.

That is the architectural move. Whether it becomes control is an empirical question.

An action input is not control

A model can receive an action and still ignore it, respond weakly, or let it change appearance without changing the underlying trajectory. Controllability has to be measured by holding the start fixed, varying only the command, and checking whether the resulting futures separate in the requested way.

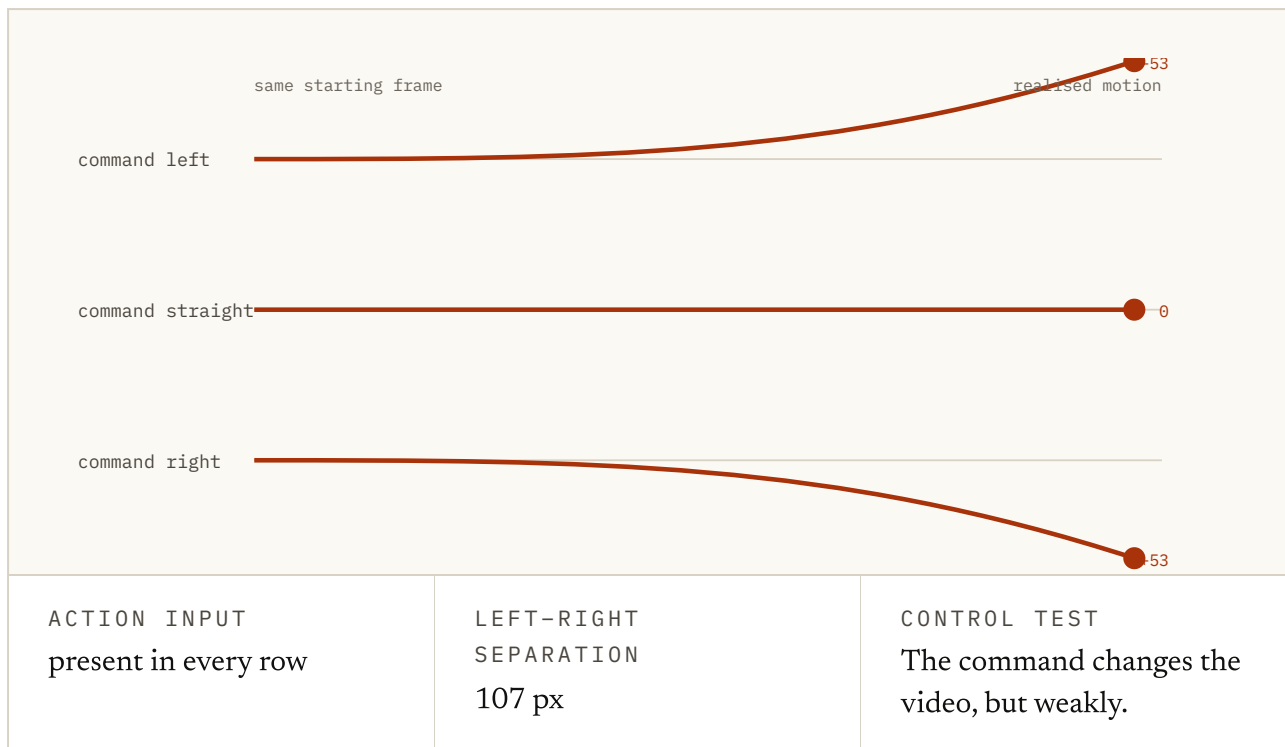


FIG. 8.2 Every row receives a different command from the same starting frame. Move action fidelity towards zero. The input remains present, but the realised paths collapse together. Conditioning describes the interface. Fidelity says whether the model honoured it.

Video quality can hide weak control. If three beautiful continuations all go straight, the system has generated video. It has not generated the counterfactuals a planner asked for.

What scaling bought

More than most people expected, including most of the people building it.

Genie 3 is reported to generate interactive video at 720p and 24 frames a second. It is reported to hold a scene together over several minutes, and to take instructions part-way through: change the weather, add a flock of birds. Walk away from something, walk back, and it is roughly as you left it.

Take those numbers as the lab's report rather than as measurements anyone outside it has repeated. That is not a slight; it is the normal state of this subject, and Chapter 9 is largely about how little of it can currently be checked.

The public interface exposes no persistent geometry or object database that you could inspect or export. That is not the same as having no memory. DeepMind says Genie 3 can refer back to its trajectory and retain visual information for about a minute. Whatever internal state supports that behaviour remains implicit in the generator rather than available as a scene graph.

What "emergent physics" means

You will see the claim made about every one of these systems. It is worth being precise about what it can and cannot mean.

The observation behind it is real. Train on enough video and the outputs start respecting things nobody put in. Dropped objects accelerate downwards. Water pools rather than piling up. Things hidden behind other things come back when the thing in front moves. No equation was supplied. Those regularities are in the training footage, and predicting the footage well requires reproducing them.

The claim built on top is where the trouble starts. "It learned physics" suggests it acquired the rule, and there is a test for that. It is not whether the outputs look right on the sort of thing it was trained on.

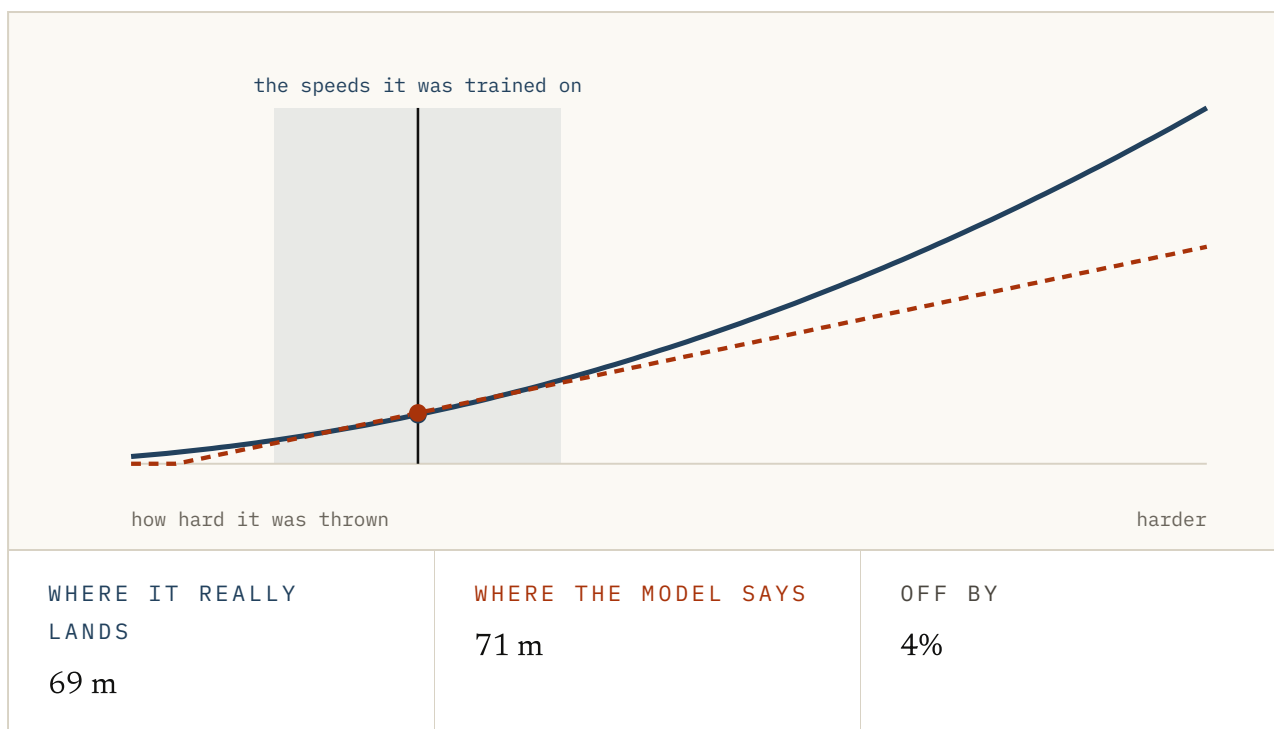


FIG. 8.3 The dashed line is a model fitted to the true behaviour across one band of speeds. Inside that band it is within a few per cent, and anyone testing it there would conclude it had the rule. Drag outside the band. It does not have the rule; it has a straight line through the part of the curve it was shown.

Everything inside the band is consistent with having learned physics. It is equally consistent with having learned the answers to the questions it was asked. The two only come apart outside, and outside is exactly where nobody tests, because outside is where you have no footage to compare against.

So the honest version of the claim is narrower and more useful. These systems reproduce a great many physical regularities across the range of situations they were trained on. Whether that constitutes having the rule is not settled by any demonstration you have seen, because the demonstrations are all inside the band.

This is not a special failing of video models. It is the ordinary problem with any fitted function, and the reason it feels different here is only that the outputs are so persuasive. A wrong answer that looks like a photograph is more convincing than a wrong number.

What they are for

Given all that, it would be easy to be dismissive, and that would be a mistake.

For training other systems, a generated world is very useful. It avoids real environment contact, but pays in accelerator time. It resets instantly and can produce a thousand variations of a scene that would take a week to stage. NVIDIA's Cosmos is built and marketed on exactly this: physically-aware generated video as training data for robots and vehicles.

For being a world you can move around in, they are already good. That is worth something on its own, and no other approach in this course delivers it.

For being the thing an agent plans inside, raw video generators are an awkward shape. A planner may need to score a hundred action sequences, and rendering each one is expensive. Systems can distil a compact state or plan in a learned latent. In that case the state does the planning work, not the visible video.

Where the seams are

Three, briefly, because the next chapter is the full accounting.

Consistency is learned, not enforced. Nothing prevents the room from changing. It usually does not, and the fact that it usually does not is a statistical property of a trained system rather than a guarantee.

The interesting moments are the vague ones. A model trained on pixels is pushed towards the average of the possible futures wherever the future is open. The future is most open exactly when something is about to happen.

The scores do not compose. Benchmarks separately measure visual quality, geometry, action fidelity and persistence. There is no accepted rule for combining those results into one ranking. A claim about improvement therefore belongs to a particular test.

Try it

1 What change makes a video model capable of being steered, without proving that it will obey?

- a. Higher resolution
- b. A longer context window
- c. More parameters
- d. Conditioning each generated frame on an action as well as on the frames before it

ANSWER d. Conditioning each generated frame on an action as well as on the frames before it. Action conditioning supplies the interface. Controllability still has to be tested by varying the action from the same start and measuring whether the requested futures separate.

2 A generated room stays as you left it when you turn back. What can you conclude about its memory?

- a. Some internal state supports persistence, but the behaviour does not reveal whether it is a scene graph, context or another mechanism
- b. It must use a separate object database
- c. It must contain exportable geometry
- d. It has no memory of any kind

ANSWER a. Some internal state supports persistence, but the behaviour does not reveal whether it is a scene graph, context or another mechanism. Behaviour establishes persistence, not implementation. Genie 3 is reported to refer back to its trajectory, while exposing no persistent geometry or object store to inspect.

3 Dropped objects in generated video accelerate downwards, and nobody supplied gravity. What does that establish?

- a. Nothing at all
- b. The model has learned the law of gravity
- c. That the regularity is in the training footage, and reproducing it is required to predict the footage well
- d. The model contains a physics engine

ANSWER c. That the regularity is in the training footage, and reproducing it is required to predict the footage well. The observation is real. The question is whether reproducing a regularity across the range you trained on amounts to having the rule, and the observation alone does not settle it.

4 In Figure 8.3, the model is within a few per cent across the band it was trained on. Why is that not evidence it has the rule?

- a. The measurement is unreliable
- b. Matching inside the band is equally consistent with having learned the answers to the questions it was asked
- c. A few per cent is too large an error
- d. The band was too narrow

ANSWER b. Matching inside the band is equally consistent with having learned the answers to the questions it was asked. Having the rule and having a fit through the sampled region only come apart outside the band, and outside is where nobody tests because there is no footage to compare against.

5 Why is a generated video world the wrong shape for an agent to plan inside?

- a. It cannot be conditioned on actions
- b. It has no reward signal
- c. It is not accurate enough
- d. Planning means running the model many times over, and a frame is the opposite of a compact state

ANSWER d. Planning means running the model many times over, and a frame is the opposite of a compact state. A planner needs to try many action sequences and score them cheaply. These models are expensive to run and produce frames rather than something small to reason over.

6 What are generated worlds unambiguously good for right now?

- a. Producing training data and being a place you can move around in
- b. Long-horizon planning
- c. Compressing video
- d. Replacing physics engines

ANSWER a. Producing training data and being a place you can move around in. They avoid real-world contact, reset instantly and produce broad variation, while still costing compute. No other approach in this course gives you an actual picture to walk into.

7 Why does a wrong answer from one of these systems feel more convincing than a wrong number?

- a. It usually is not wrong
- b. The errors are smaller
- c. It arrives as something that looks like a photograph
- d. The models are better calibrated

ANSWER c. It arrives as something that looks like a photograph. The extrapolation problem here is the ordinary problem with any fitted function. What is different is how persuasive the output is when it fails.

8 A system improves its geometry score but loses action fidelity. Is it better overall?

- a. Yes, geometry is the only objective measure

- b. Not without a declared use case or rule for combining the dimensions
- c. No, action fidelity always dominates
- d. Yes, any benchmark gain establishes overall progress

ANSWER b. Not without a declared use case or rule for combining the dimensions. Benchmarks now measure several useful dimensions. The unresolved part is how to combine them when systems and applications value different contracts.

FIG. 8.4 Eight questions on the move and the claim. The middle three are the ones worth being careful about when a demo is impressive.

Which brings us to the part of the subject that is usually left out of the announcements.

Thanks for reading. Chapter 9 is the honest accounting: what drifts, what gets forgotten, what cannot be asked, and why nobody can currently tell you which of these systems is best.

Sources

Genie: Generative Interactive Environments BRUCE ET AL., 2024 ↗

Latent actions learned from unlabelled video, which is the move that turns a video model into somewhere you can be.

<https://arxiv.org/abs/2402.15391>

Genie 3 GOOGLE DEEPMIND, 2025 ↗

The reported numbers this chapter quotes: 720p, 24 frames a second, minutes of coherence. A lab report rather than a result anyone outside has repeated.

<https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>

Diffusion Models Are Real-Time Game Engines (GameNGen) VALEVSKI ET AL., 2024 ↗

DOOM generated frame by frame from previous frames and inputs, fast enough to play. The clearest demonstration that this is playable rather than merely watchable.

<https://arxiv.org/abs/2408.14837>

How Far is Video Generation from World Model: A Physical Law Perspective

KANG ET AL., 2024 ↗

The measured version of Figure 8.3. Video models match physical laws within the distribution they were trained on, and do not extrapolate outside it.

<https://arxiv.org/abs/2411.02385>

Video generation models as world simulators OPENAI, 2024 ↗

The report that made the emergent-physics claim a mainstream one. Worth reading for exactly what it does and does not assert.

<https://openai.com/index/video-generation-models-as-world-simulators/>

Cosmos World Foundation Model Platform for Physical AI NVIDIA, 2025 ↗

Generated video built as training data for robots and vehicles, which is the use these systems are unambiguously good for.

<https://arxiv.org/abs/2501.03575>

Cosmos NVIDIA ↗

The product framing, and a useful boundary case: generative video beside explicit simulation, sold as one platform.

<https://www.nvidia.com/en-us/ai/cosmos/>

FIG. 8.5 Ordered for someone building on this rather than for historical completeness.